

Special PDFs

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Following with Python

Download the notebook from the canvas page for this week's material:
[SpecialPDF.py](#)

```
marimo edit SpecialPDF.py
```

Special PDFs

- Intro
- Bernoulli
- Binomial
- Poisson
- Uniform
- Exponential
- Beta
- Normal aka Gaussian

Intro, reminder of big picture

If a RV follows a special distribution, we can immediately use it to calculate the probabilities eg, $P(X \leq x)$ for $X \sim \text{Poisson}(\lambda)$.

BUT we need to know the parameter(s) of the special distribution to do this. If the parameters are not known, we have to estimate them using data. This is the whole thing we are working towards.

We will use this week to:

Figure out which PDF is for what

Practice calculating Probabilities, expectation, variance

Bernoulli

The possible outcomes of **a single coin toss** experiment are described by a Discrete RV K , and K follows a Bernoulli PMF.

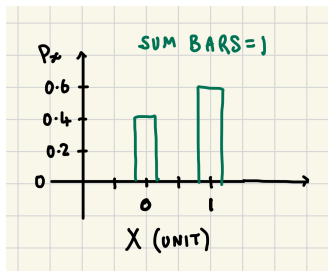
$$K \sim \text{Bernoulli}(p)$$

The parameter is p : the probability of (eg) success.

Bernoulli is the distribution we use to describe the outcome of a single experiment with two possible outcomes: success/failure; heads/tails; on/off;

Bernoulli PMF

Recall our PMF sketch from week 2 with two bars, one with $p_k = p$, one with $p_k = 1 - p$:

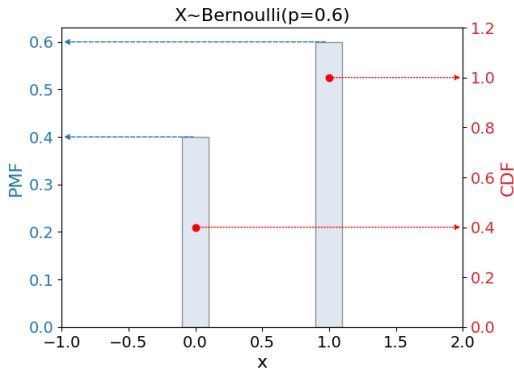


Our PMF is:

$$p_K(k) = \begin{cases} 0 & k < 0 \\ 1 - p & k = 0 \\ p & k = 1 \\ 0 & k > 1 \end{cases}$$

The support of K is all values k with non-zero probability: $[0,1]$. We write down the PMF for the support and for 'outside the support' for completeness.

Bernoulli PMF Plot



The CDF is not super helpful in this case, but will be useful later.

Bernoulli PMF: validity

Is $\sum_{j_i \in S} p_K(k_i) = 1$?

$$p_K(k) = \begin{cases} 0 & k < 0 \\ 1 - p & k = 0 \\ p & k = 1 \\ 0 & k > 1 \end{cases}$$

Normalisation check: $\sum_{k_i \in S} p_K(k_i) = p + (1 - p) = 1$: the Bernoulli PMF is valid.

Bernoulli Probabilities

The CDF $F_K(k) = P(K \leq k) = \sum_{k_i}^k p_K(k_i)$ must be written in terms of ranges of values, because the Probability from the CDF is always for a range.

$$p_K(k) = \begin{cases} 0 & k < 0 \\ 1 - p & k = 0 \\ p & k = 1 \\ 0 & k > 1 \end{cases}$$

$$F_K(k) = \begin{cases} 0 & k < 0 \\ 1 - p & 0 \leq k < 1 \\ 1 & 1 \leq k \end{cases}$$

Bernoulli: Expected Value & Variance

Expected Value:

$$E[K] = \sum_{k_i \in S} k_i p_K(k_i) = 0(1 - p) + 1(p) = p$$

Variance:

$$V[K] = E[K^2] - E^2[K]$$

$$E[K^2] = \sum_{k_i \in S} k_i^2 p_K(k_i) = 0^2(1 - p) + 1^2(p) = p$$

$$E^2[K] = p^2$$

$$V[K] = p - p^2 = p(1 - p)$$

Bernoulli: python

```
from scipy.stats import bernoulli  
bernoulli.pmf(k, p, loc=0)
```

k is a number of successes, or user-provided array of values for the x-axis.

p is the parameter: probability of success

loc=0 is the default value of the location parameter, allowing for sliding along the x-axis if wanted. For example if for some reason you wanted success=6 and failure=5, you could set loc=5

Binomial

This time we will **toss the coin n times**. I will count the number of heads, k , and that count is my Discrete RV K .

$$K \sim \text{Binomial}(n, p)$$

The parameters are n, p where p is again the probability of (eg) success.

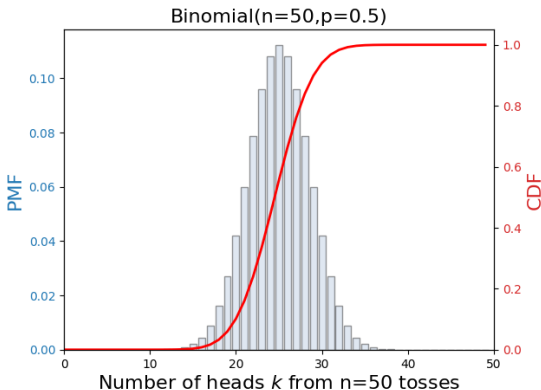
Note that one Binomial trial consists of n Bernoulli trials: Bernoulli was the number of heads from one toss.

Binomial PMF

The PMF is a bit of a mouthful:

$$p_K(k) = \begin{cases} 0 & k < 0 \\ C_k^n p^k (1-p)^{n-k} & 0 \leq k \leq n \\ 0 & k > n \end{cases}$$

Binomial PMF plot



This is theory, not data. To generate this distribution with actual coin tosses you would get one data point for each set of 50 tosses. Then you could repeat it maybe a thousand times to get a distribution something like this.

Binomial Coefficient

$$C_k^n = n! / k!(n - k)! : (\text{N combinations}) / (\text{N successful})(\text{N unsuccessful})$$

For example, $C_3^4 = 4! / 3!(4 - 3)! = 4 \cdot 3 \cdot 2 \cdot 1 / [(3 \cdot 2 \cdot 1) \cdot (1)] = 4$

Factorials:

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$$3! = 3 \cdot 2 \cdot 1 = 6$$

$$2! = 2 \cdot 1 = 2$$

$$1! = 1$$

$$0! = 1 \text{ by definition}$$

$$-1! = \text{NAN undefined}$$

$$a! / (a-1)! = a$$

Binomial Coefficients can be very large

$$C_k^n = n! / k!(n - k)! : (N \text{ combinations}) / (N \text{ successful})(N \text{ unsuccessful})$$

Binomial coefficients can be **very large**.

$a!/(a-1)! = a$: this is useful only when we have a single 'failure'.

For example if we toss a coin 20 times and get 9 heads, the number of ways that can happen is:

```
import math
# 20 tosses, 9 heads
math.factorial(20)/ ( math.factorial(9)*math.factorial(11) )
167_960
# 30 tosses, 15 heads
math.factorial(30)/ ( math.factorial(15)*math.factorial(15) )
155_117_520
```

Binomial PMF: validity

$$p_K(k) = C_k^n p^k (1-p)^{n-k} \text{ for } 0 \leq k \leq n$$

$$\text{Is } \sum_{j_i \in S} p_K(k_i) = 1 ?$$

There is no predefined limit on n , and k can take any non-negative value up to and including n : we have a sum to infinity. When that happens we need a SERIES. The series we need in this case is called the binomial series (or binomial theorem):

$$(a + b)^n = \sum_{k=0}^{\infty} C_k^n a^k b^{n-k}: \text{ we have } a = p, b = 1 - p$$

Normalisation check: $\sum_{k=0}^{\infty} p_K(k_i) = (p + (1 - p))^n = 1^n = 1$: the Binomial PMF is valid.

Binomial Probabilities

The Probability of $k \leq m$ successes is:

$$P(K \leq m) = \sum_{k=0}^m C_k^n p^k (1-p)^{n-k}$$

For example, the probability of tossing a coin $n=5$ times and getting **up to** $k=4$ heads on the toss of a fair coin with $p = 0.5$ is:

$$P(K \leq 4) = \sum_{k=0}^4 \frac{5!}{k!(5-k)!} 0.5^k 0.5^{5-k} = 0.96875$$

The probability of getting **at least** $k=4$ heads is calculated as:

$$1 - P(K \leq 3) = 1 - \sum_{k=0}^3 \frac{5!}{k!(5-k)!} 0.5^k 0.5^{5-k} = 0.1875$$

Binomial: Expected Value

Expected Value:

$$E[K] = \sum_{k_i \in S} k_i p_K(k_i)$$

Note that the total number of successes $k = \sum k_i$, so we can write:

$$E[K] = E \left[\sum_{k_i \in S} k_i \right] = \sum_{k_i \in S} E[k_i]$$

The expectation of a sum is the sum of expectations: $E[X + Y] = E[X] + E[Y]$

$E[k_i]$ is the probability of success, p . So $E[K] = \sum_0^n p = np$

Binomial: Variance

Variance:

$V[K] = E[K^2] - E^2[K]$: for once we will not use this form, as it is a long road.

Because Binomial is n repeated Bernoulli trials, we can extract the variance directly from *Bernoulli* $V[K] = p(1 - p)$.

Binomial $V[K] = np(1-p)$.

The variance of a sum of **independent** experiments is the sum of the variance of each of them. $V[\text{sum}] = \text{sum}[V]$

Binomial: python

```
from scipy.stats import binom  
binom.pmf(k, n, p, loc=0)
```

k is number of successes, or user-provided array of values for the x-axis.

n is the parameter: number of trials

p is the parameter: probability

loc=0 is the location parameter

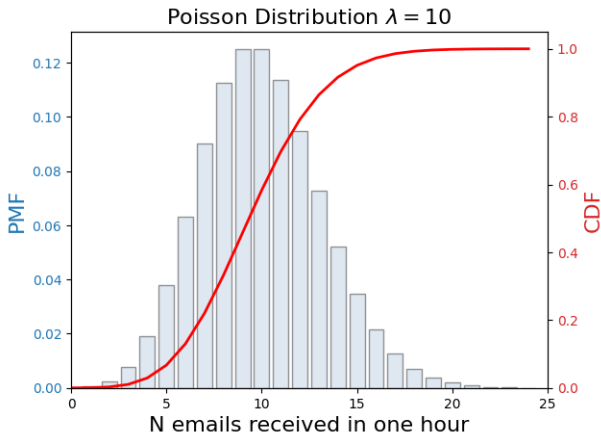
Poisson

We don't just want to model coin flips. Something very important to humans is time. What is the count of 'successful' outcomes **in a given time period**?

$$K \sim \text{Poisson}(\lambda)$$

The parameter is λ : the success rate.

Poisson PMF plot



Poisson PMF: validity

$$\text{Is } \sum_{n=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} = 1?$$

For this PMF we again have an infinite range for our RV. When that happens we need a SERIES. We can look up the exponential series :

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}: \text{ good for us with } x \rightarrow \lambda \text{ and } n \rightarrow k$$

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = e^{-\lambda+\lambda} = e^0 = 1 \text{ because } e \text{ is a number.}$$

Normalisation check: the Poisson PMF is valid.

Poisson Probabilities

The CDF is as usual: $P(K \leq k) = F_K(k) = \sum_{k_i=0}^k p_K(k_i)$

The PMF is $p_K(k) = e^{-\lambda} \frac{\lambda^k}{k!}$ for $k \geq 0$

I receive $\lambda \approx 100$ emails per 10-hour day. What is the probability of me getting zero emails in a given hour?

$\lambda = 10$ is the rate per hour.

$$P(K = 0) = F_K(0) = e^{-10} \frac{10^0}{0!} = e^{-10} \frac{1}{1} \approx 4.5 \times 10^{-5}$$

Poisson: Expected Value

Expected Value:

$$E[K] = \sum_{k_i=0}^{\infty} k_i p_K(k_i) = \sum_{k_i=0}^{\infty} k_i e^{-\lambda} \frac{\lambda^{k_i}}{k_i!}$$

I want to use the exponential series to make this sum manageable.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

It is going to need massaging into shape to have the right form.

Note: $\frac{k}{k!} = \frac{1}{(k-1)!}$

Also: $\lambda^k = \lambda^{k-1+1} = \lambda^{k-1}\lambda$

Poisson: Expected Value

Expected Value:

$$E[K] = \sum_{k_i=0}^{\infty} k_i p_K(k_i) = \sum_{k_i=0}^{\infty} k_i e^{-\lambda} \frac{\lambda^{k_i}}{k_i!}$$

Note 2: $\frac{k}{k!} = \frac{1}{(k-1)!}$ and $\lambda^k = \lambda^{k-1+1} = \lambda^{k-1}\lambda$

$$E[K] = e^{-\lambda} \sum_{k_i=0}^{\infty} \frac{\lambda^{k_i}}{(k_i-1)!} = \lambda e^{-\lambda} \sum_{k_i=0}^{\infty} \frac{\lambda^{k_i-1}}{(k_i-1)!}$$

Now we see we can use the exponential series again:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}: \text{ good for us with } x \rightarrow \lambda \text{ and } n \rightarrow k-1$$

$$E[K] = \lambda e^{-\lambda} e^{\lambda} = \lambda: \text{ which makes sense.}$$

Note: its okay that we have the $n \rightarrow k-1$ in the sum, because the $k=0$ term is zero.

Poisson: Variance

Variance:

$$V[K] = E[K^2] - E^2[K] = \lambda$$

OPTIONAL! Proof of this is a little bit fiddly but not too long. Give it a go, or/and see proof here: [proofwiki](#).

Don't ~~waste~~ spend time trying this unless everything we are doing is a total cake-walk, and you are an expert at python, and you know how to make beautiful plots.

Poisson: python

```
from scipy.stats import poisson
poisson.pmf(k, mu, loc=0)
```

k is a number per time period, or user-provided array of values for the x -axis.

μ is the parameter we have called λ . This makes sense because the expected value $E[K] = \lambda$, and we use the notation $E[K] \equiv \mu$.

$\text{loc}=0$ is the location parameter

Note: λ has a special meaning in python, so is never used for a parameter name.

Poisson versus Binomial

My friend Alf owns a donut shop in Brighton. On average, he sells 17.4 donuts a day (he is only open between 09:30 and 10:30).

How should he model the number of donuts he sells as an RV?



Poisson versus Binomial

My friend Alf owns a donut shop in Brighton. On average, he sells 17.4 donuts a day (he is only open between 09:30 and 10:30).

How should he model the number of donuts he sells as an RV?

The obvious choice is to go with
 $X \sim \text{Poisson}(\lambda = 17.4)$.

A less obvious choice would be to go with $X \sim \text{Binomial}(n, p)$: n is the number of people in Brighton, p is the probability that any of them buys a donut.



Poisson versus Binomial

Comparing the expectations, Poisson $E[K] = \lambda$, Binomial $E[K] = np$. So $\lambda = np$ or $p = \lambda/n$.

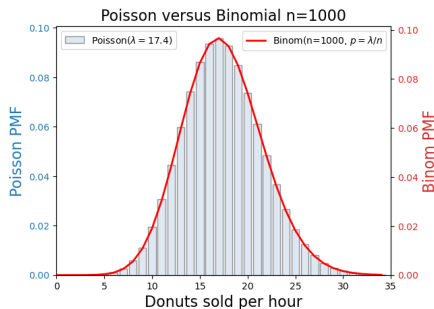
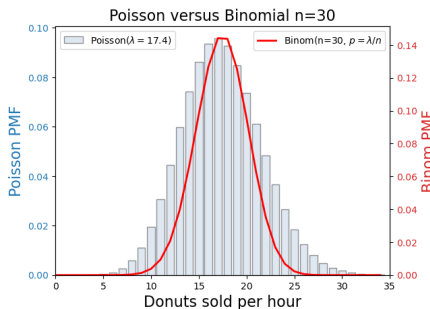
In the limit of $n \rightarrow \infty$, the Binomial is a valid model. Note that the rate λ has no limit on the number of people it draws from.



In the limit of $n \rightarrow \infty$, $\text{Binomial}(n, \lambda/n) \rightarrow \text{Poisson}(\lambda)$

Alf's donut shop

Alf concludes that it is safer to use $X \sim \text{Poisson}(\lambda)$ as his model for donut sales.



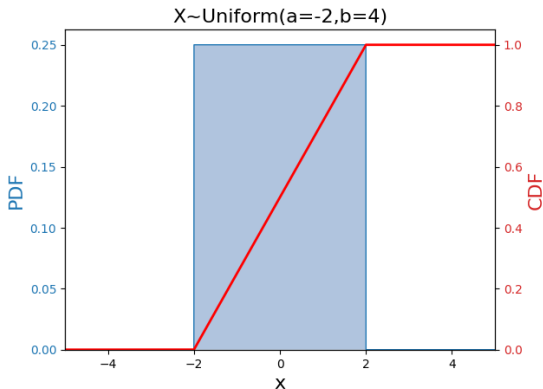
Now: Special PDFs (continuous RVs)

Uniform PDF

We can write the PDF:

$$f_X(x) = \begin{cases} \text{constant} & a \leq x \leq b \\ 0 & \text{elsewhere} \end{cases}$$

Uniform PDF Plot



Note! the values on the y-axis are meaningless for PDFs. Cannot read off individual probabilities like we can for PMFs.

Uniform PDF: validity

We have so far only defined the PDF as $f_X(x) = c$ for $a \leq x \leq b$, so this is very easy.

$$\int_a^b c \, dx = c x \Big|_a^b = c(b - a) = 1, \text{ so } c = \frac{1}{b - a}$$

Normalisation check gives us the form $f_X(x) = \frac{1}{b - a}$ for $a \leq x \leq b$

The Uniform PDF is valid.

Uniform Probabilities

Recall that we can *only* use the CDF (not the PDF) for probabilities associated with CRVs.

$$P(X \leq x) = F_X(x) = \int_a^x \frac{1}{b-a} dx = \frac{x}{b-a} \Big|_a^x = \frac{x-a}{b-a}$$

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

Uniform PDF Expected Value & Variance

Expected Value:

$$E[X] = \int_a^b x f_X dx = \int_a^b \frac{x}{b-a} dx = \frac{x^2}{2(b-a)} \Big|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{(b+a)}{2}$$

Variance:

$$V[X] = E[X^2] - E^2[X]$$

$$E^2[X] = \frac{(b+a)^2}{4}$$

$$E[X^2] = \int_a^b \frac{x^2}{b-a} dx = \frac{b^3 - a^3}{3(b-a)}$$

$$V[X] = \frac{b^3 - a^3}{3(b-a)} - \frac{(b+a)^2}{4} = \frac{(b-a)^2}{12}$$

optional algebra exercise: prove this.

Uniform PDF: python

```
from scipy.stats import uniform
uniform.pdf(x, loc=0, scale=1)
```

x is a user-provided range of values for the x -axis.

$loc=0$ is the location parameter. For us this is $loc=a$

$scale=1$ is the scale parameter. For us this is $scale=b-a$

Exponential

The exponential distribution is essentially a 'waiting times' distribution.

$$T \sim \text{Expon}(\lambda)$$

I am using $X \rightarrow T$ for the single RV because I want us to be really clear that this is time. I am a physicist...

Exponential PDF

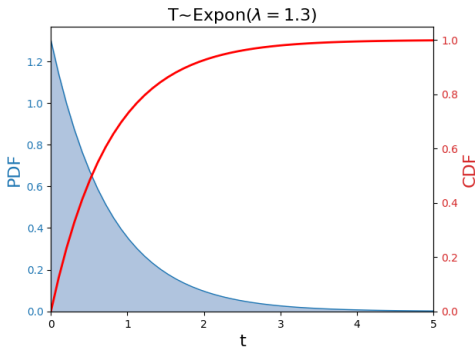
$$f_T(t) = \lambda e^{-\lambda t} \text{ for } 0 \leq t \leq \infty$$

Compare with the Poisson PDF which models K : counts in a given time.

Recall, $K \sim \text{Poisson}(\lambda) : p_K(k) = \frac{\lambda^k}{k!} e^{-\lambda}$

In both cases we have $\lambda = \text{events}/\text{time}$ but for Poisson our expectation is a count for fixed time , and we shall see that for Expon the expectation is a time for fixed counts.

Exponential PDF Plot



It may not look like it, but the CDF does not actually reach 1 until $t = \infty$.

Exponential PDF Validity

$$\text{Is } \int_0^{\infty} \lambda e^{-\lambda t} dt = 1?$$

$$\int_0^{\infty} \lambda e^{-\lambda t} dt = -\frac{\lambda}{\lambda} e^{-\lambda t} \Big|_0^{\infty} = -[e^{-\infty} - e^0] = -[0 - 1] = 1$$

The Exponential PDF is valid for $0 \leq t \leq \infty$.

Exponential Probabilities

$$P(T \leq t) = F_T(t) = \int_0^t \lambda e^{-\lambda t} dt = -[e^{-\lambda t} - e^0] = 1 - e^{-\lambda t}$$

I can use this to figure out the probability for example that I will wait less than or equal to one hour to hitch a lift on a road with an average rate of hitch-hiker pickups of $\lambda = 1.3$ per hour.

$$P(T \leq 1H) = 1 - e^{-1.3} = 0.727$$

What would Poisson give us? Now we are asking the question: 'what is the probability that a count of not zero occurs in a time period of 1H?'

$$p_K(k=0) = e^{-1.3} \frac{1.3^0}{0!} = e^{-1.3} = 0.273 : \text{probability of zero pickups.}$$

Probability of 'not zero' in 1H is $1 - e^{-1.3} = 0.727$ as before.

Exponential Expected Value

Expected Value:

$$E[T] = \int_0^{\infty} t f_T dt = \int_0^{\infty} t \lambda e^{-\lambda t} dt$$

This integral is a product of two terms in t , so we have to do it piecewise (integration by parts). This is an excellent exercise, try it if you have time.

I did: $u = t$, $u' = 1$, $v = -e^{-\lambda t}$, $v' = \lambda e^{-\lambda t}$

You will find $E[T] = \frac{1}{\lambda}$

Exponential PDF: python

```
from scipy.stats import expon
expon.pdf(x, loc=0, scale=1)
```

x is a user-provided range of (time) values for the x-axis.

$loc=0$ is the location parameter.

$scale=1$ is the scale parameter. The scale is $1/rate=1/\lambda$.

Exponential PDF is **Memoryless**

Let $A = T \geq t + n$ and $B = T \geq n$. Note that $B \subset A$.

This means that A is the event where the wait is longer than $t+n$. This event A entirely contains the event where the wait is longer than t , for all non-negative values of n .

Let's remind ourselves of the exponential PDF and CDF:

$$f_T(t) = \lambda e^{-\lambda t} \text{ for } 0 \leq t \leq \infty$$

$$P(T \leq t) = F_T(t) = \int_0^t \lambda e^{-\lambda t} dt = -[e^{-\lambda t} - e^0] = 1 - e^{-\lambda t}$$

$$P(T \geq t) = 1 - F_T(t) = e^{-\lambda t}$$

We are not double counting $t = 0$, because $P(t = 0) = 0$ for a CRV.

$$P(A) \equiv P(T \geq t + n) = 1 - P(T \leq t + n) = e^{-\lambda(t+n)}.$$

$$P(B) \equiv P(T \geq n) = 1 - P(T \leq n) = e^{-\lambda(n)}.$$

Exponential PDF is **Memoryless**

Let $A = T \geq t + n$ and $B = T \geq n$. Note that $B \subset A$.

$$P(A) \equiv P(T \geq t + n) = 1 - P(T \leq t + n) = e^{-\lambda(t+n)}.$$

$$P(B) \equiv P(T \geq n) = 1 - P(T \leq n) = e^{-\lambda(n)}.$$

Conditional probability $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$ because $B \subset A$.

$$P(A|B) = e^{-\lambda(t+n)} / e^{-\lambda n} = e^{-\lambda(t+n-n)} = e^{-\lambda t}.$$

What this tells us is the probability of our new waiting time is independent of the time we have already waited. Bah!

Beta PDF

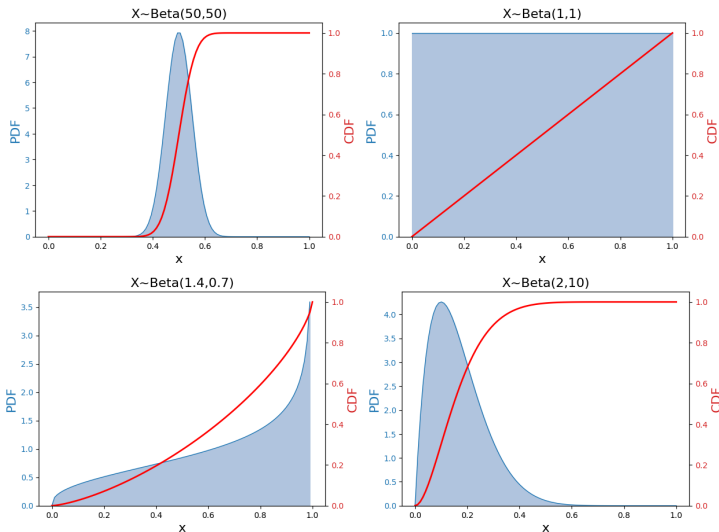
$$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \text{ for } 0 \leq x \leq 1, \alpha > 0, \beta > 0$$

The limits on the RV: $0 \leq x \leq 1$: **we can use this when the RV is itself a probability.**

If we don't know exactly what the probability is, but have some idea of the expectation and variance, we can use the Beta distribution .

Normalisation factor: $\frac{1}{B(\alpha, \beta)} = \frac{1}{\int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx}$.

Beta PDF Plots: The Chameleon!



Beta Variance & Expectation

Integrating the Beta PDF is a bit too much like hard work for me, but I trust my sources for these:

$$E[X] = \int_0^1 x f_X dx = \frac{\alpha}{\alpha + \beta}$$

$$V[X] = E[X^2] - E^2[X] = \frac{E[X](1 - E[X])}{\alpha + \beta + 1} = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

I recommend (again?) looking at the [Probability Playground](#).

Beta PDF Example

We do not know if a coin is fair (hypothesis $p = 0.5$) but we are studying it. We would like to generate a prior pdf with $E[X] = 0.5$ and standard deviation $\sigma = \sqrt{V[X]} = 0.1$. We can use a Beta distribution!

Want $E[X] = 0.5$, so $E[X] = \frac{\alpha}{\alpha + \beta}$ means we must have $\alpha = \beta$.

Want $V[X] = 0.1^2$, so $V[X] = \frac{0.5(1 - 0.5)}{\alpha + \beta + 1} = \frac{0.25}{2\alpha + 1} \therefore \alpha = 12 (= \beta)$

We will use $\pi(\theta) = \text{Beta}(12, 12) = \frac{1}{B(a, b)} x^{11} (1 - x)^{11}$

When we look at this again later you will see how useful it is for Bayesian stats, but this is just a taster for now.

Normal aka Gaussian

The Normal distribution is the **Queen Of Them All**.

$$X \sim \text{Norm}(\mu, \sigma)$$

Notice that the parameters are μ , σ : these are the symbols we use to express the expectation and standard deviation of any distribution.

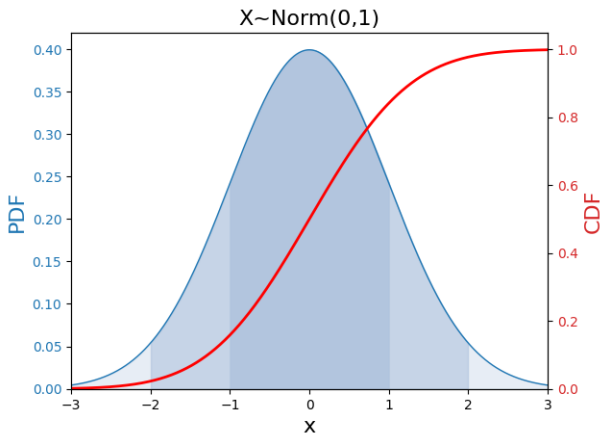
Normal PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Often written in **Standardised** form with $Z = \frac{X - \mu}{\sigma}$:

$$f_Z(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} z^2}$$

Normal PDF Plot



Normal PDF Validity

Is $\int_{-\infty}^{\infty} f_X(x) dx = 1$? Yes :)

Integral is $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$

We can note the result: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

See this super-clear proof from [Christine Breiner](#) (youtube).

This gives us the normalisation factor of $\frac{1}{\sqrt{2\pi}}$ in the PDF.

Normal Probabilities

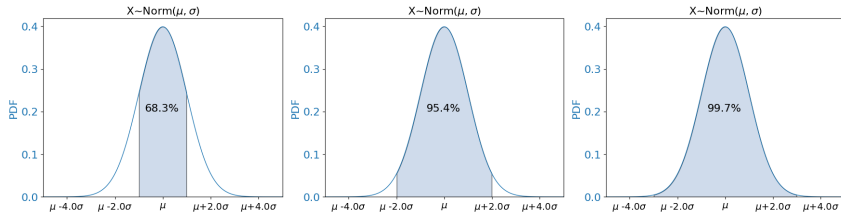
There is no analytical solution to the indefinite integral of the Gaussian PDF.

Rather than calculating the CDF to get probabilities, we tend to look them up (some other kind soul did the approximations for us and made lookup tables). I would use python, which does the same (error function) calculations used for the lookup tables.

We can also use the very special properties of the Normal distribution to note down some useful probabilities.

Normal Probabilities

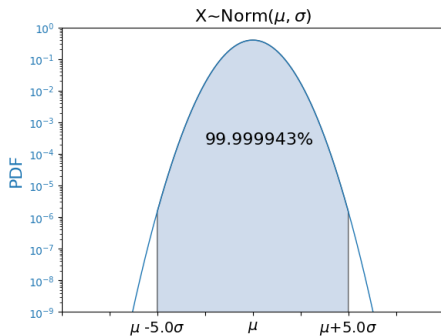
No matter what values of μ and σ , the normal distribution will always have 68%, 95%, 99.7% of its area within $\pm 1, 2, 3$ standard deviations from the mean.



You would be forgiven for thinking that this distribution has its limits around $\pm 4\sigma$ from the mean. But the Normal distribution has limits of $\pm \infty$.

Particle Physicist's version

To see that the Gaussian distribution has tails (out to $\pm\infty$, but we can't plot that) we need to zoom on the lower part of the y-axis, which we can do by using a **log scale**. We also need more significant figures. Particle physicists don't use the D-word unless we see something outside the 5σ range of probabilities, corresponding to a p-value of 5.7×10^{-7} (a one in 5.7 million chance that it is a statistical fluctuation rather than an actual Discovery) .



Normal PDF: python

```
from scipy.stats import norm
norm.pdf(x, loc=0, scale=1)
```

x is a user-provided range of values for the x -axis.

$loc = \mu = E[X]$

$scale = \sigma = \sqrt{V[X]}$

Loads more coming on the Gaussian.

If you thought it was weird that the $\text{Binomial}(n, p)$ is $\text{Poisson}(\lambda)$ in the limit of $n \rightarrow \infty$, just wait until you see what the Gaussian has up its sleeves.

Before we go any further with the Gaussian, **we need to talk about correlations and uncertainties.**

Thanks For Your Time

Fin.